Database Joins

Database Management Systems

SQL Joins

- A JOIN clause is used to combine rows from two or more tables, based on a related column between them.
- Consider the two tables in next slide

SQL Joins

- Orders

OrderID	CustomerID	OrderDate
10308	2	1996-09-18
10309	37	1996-09-19
10310	77	1996-09-20

- Customers

CustomerID	CustomerName	ContactName	Country
1	Alfreds Futterkiste	Maria Anders	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mexico

SQL Joins

- Notice that the "CustomerID" column in the "Orders" table refers to the "CustomerID" in the "Customers" table.
- The relationship between the two tables above is the "CustomerID" column.

Joins Types in SQL

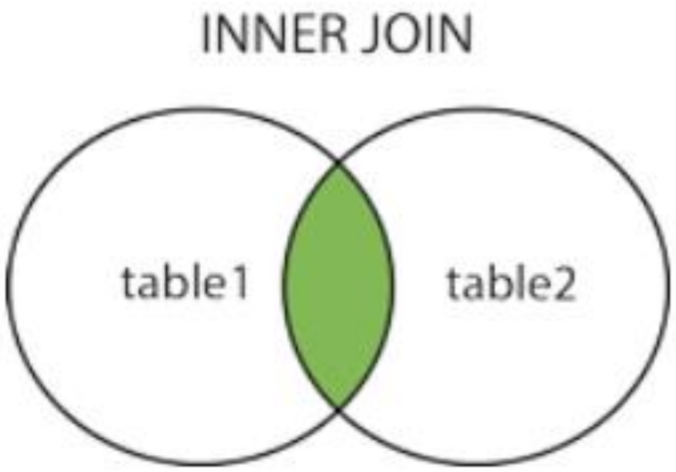
- **INNER JOIN**: Returns records that have matching values in both tables
- **LEFT JOIN**: Returns all records from the left table, and the matched records from the right table
- **RIGHT JOIN**: Returns all records from the right table, and the matched records from the left table
- **CROSS JOIN(Full Outer Join):** Returns all records from both tables

SQL Inner Join

- The INNER JOIN keyword selects records that have matching values in both tables.
- The INNER JOIN selects all rows from both participating tables as long as there is a match between the columns.
- Syntax

```
Select …. From Table1
Inner Join Table2
ON Table1.column_name = Table2.Column_name
```

SQL Inner Join



Example

Orders

OrderID	CustomerID	EmployeeID	OrderDate	ShipperID
10308	2	7	1996-09-18	3
10309	37	3	1996-09-19	1
10310	77	8	1996-09-20	2

Customers

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico

Example

- ```sql
  SELECT Orders.OrderID, Customers.CustomerName, Orders.OrderDate
  FROM Orders
  INNER JOIN Customers ON Orders.CustomerID = Customers.CustomerID;
  ```

OrderID	CustomerName	OrderDate
10308	Ana Trujillo Emparedados y helados	9/18/1996
10365	Antonio Moreno Taquería	11/27/1996
10383	Around the Horn	12/16/1996
10355	Around the Horn	11/15/1996
10278	Berglunds snabbköp	8/12/1996
  ```

Inner Join with 3 tables

Customers

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden

Orders

OrderID	CustomerID	EmployeeID	OrderDate	ShipperID
10248	90	5	1996-07-04	3
10249	81	6	1996-07-05	1
10250	34	4	1996-07-08	2
10251	84	3	1996-07-08	1
10252	76	4	1996-07-09	2

- Shippers

ShipperID	ShipperName	Phone
1	Speedy Express	(503) 555-9831
2	United Package	(503) 555-3199
3	Federal Shipping	(503) 555-9931

Example

SELECT Orders.OrderID, Customers.CustomerName, Shippers.ShipperName

FROM ((Orders INNER JOIN Customers ON Orders.CustomerID = Customers.CustomerID)

INNER JOIN Shippers **ON** Orders.ShipperID = Shippers.ShipperID);

Self Join in SQL

- A self join is a regular join, but the table is joined with itself.
- A self-join is a join that can be used to join a table with itself. Hence, it is a unary relation.
- In a self-join, each row of the table is joined with itself and all the other rows of the same table. Thus, a self-join is mainly used to combine and compare the rows of the same table in the database.
- Syntax

```
SELECT column_name(s)
FROM table1 T1, table1 T2
WHERE condition;
```

Example

- Consider the table Students

id	name	city	department
1	Mohan	Pune	CSE
2	Roy	Bangalore	CSE
3	Harry	Hyderabad	ME
4	Ritesh	Pune	ME
5	Suresh	Bangalore	CSE

- If we want to take students of same city we can apply self join on that

Example

```
SELECT s1.name AS student1, s2.name AS student2,
s1.city AS city
FROM student AS s1, student AS s2
WHERE s1.city = s2.city AND s1.id <> s2.id;
```